Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

Find the length of the chord, along the x-axis, of the circle which is orthogonal to the three circles $x^2+y^2-2x+3y-7=0, x^2+y^2+5x-5y+9=0$ and $x^2+y^2+7x-9y+29=0$

(A) $4\sqrt{17}$

(B) $2\sqrt{17}$

(C) $\sqrt{17}$

(D) $3\sqrt{17}$

Q.2

Find the coordinates of the common points of the circles $x^2+y^2-6x-6y-7=0$ and $x^2+y^2-10x+7=0$ if they intersect.

(A) $(8,3);(14/13,-21/13)$

(B) $(4,3);(14/13,-21/13)$

(C) $(8,6);(14/13,-21/13)$

(D) $(4,3);(7/13,-21/13)$.

Q.3

The range of value of r for which the point $\left(-5+\frac{r}{\sqrt{2}},-3+\frac{r}{\sqrt{2}}\right)$ is an interior point of the major segment of the circle $x^2+y^2=16$, cut-off by the line $x+y=2$, is $-$

(A) $(-\infty,5\sqrt{2})$

(B) $(4\sqrt{2}-\sqrt{14},5\sqrt{2})$

(C) $(4\sqrt{2}-\sqrt{14},4\sqrt{2}+\sqrt{14})$

(D) None of these

Q.4

The equations of the sides $AB,\ BC$ and CA of a ΔABC are $x+y=1,\ 4x-y+4=0$ and $2x+3y=6$. Circles are drawn on $AB,\ BC$ and CA as diameter. The point of concurrence of the common chord is

(A) centroid of the triangle

(B) orthocenter

(C) circumcentre

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(D) incentre

Q.5

The locus of the centre of a circle which touches extermally the circle $x^2 + y^2 - 6x - 6y + 14 = 0$ and also touches the y-axis is

(A) $x^2 - 6x - 10y + 14 = 0$

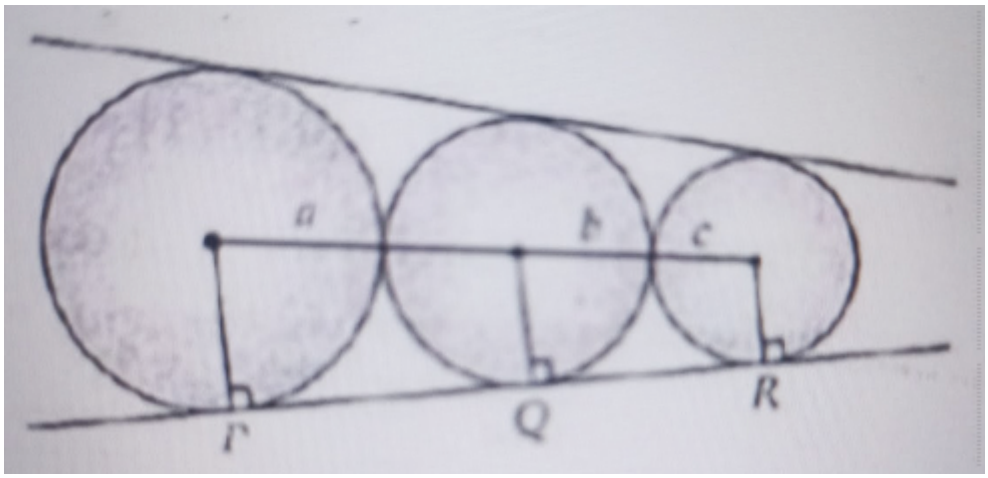
(B) $x^2 - 10x - 6y + 14 = 0$

(C) $y^2 - 6x - 10y + 14 = 0$

(D) $y^2 - 10x - 6y + 14 = 0$

Q.6

Three circles of radii a, b, c touch as shown and have common tangents. Then $b =$



(A) $\frac{a+c}{2}$

(B) $\sqrt{ac}$

(C) $\frac{2ac}{a+c}$

(D) $\frac{ac}{a+c}$.

Q.7

Find the equations of the common tangents to the circles $x^2 + y^2 - 2x - 6y + 9 = 0$ and $x^2 + y^2 + 6x - 2y + 1 = 0$

(A) $\mathrm{y} = 4; 4\mathrm{x} - 3\mathrm{y} = 0$

(B) $\mathrm{y} = 3; 4\mathrm{x} - 3\mathrm{y} = 0$

(C) $\mathrm{y} = 4; 4\mathrm{x} + 3\mathrm{y} = 0$

(D) $\mathrm{y} = 4; 3\mathrm{x} - 4\mathrm{y} = 0$

Q.8

The equation of the circle, which is touched by the line $y = x$, has its centre on the positive x-axis. Find $\frac{r^2}{2}$ (where r is radius of the circle).

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(A) 1

(B) 2

(C) 0.5

(D) 4

Q.9

Consider two circles $C_1 : x^2 + y^2 - 1 = 0$ and $C_2 : x^2 + y^2 - 2 = 0$. Let $A(1, 0)$ be a fixed point on the circle C_1 and B be any variable point on the circle C_2. The line BA meets the curve C_2 again at C. Which of the following alternative(s) is/are correct?

(A) $\mathrm{OA}^2 + \mathrm{OB}^2 + \mathrm{BC}^2 \in [7, 11]$, where O is the origin

(B) $\mathrm{OA}^2 + \mathrm{OB}^2 + \mathrm{BC}^2 \in [4, 7]$, where O is the origin.

(C) Locus of midpoint of AB is a circle of radius $\frac{1}{\sqrt{2}}$

(D) Locus of midpoint of AB is a circle of area $\frac{\pi}{2}$

Q.10

Find the equation of chord of the circle $x^2 + y^2 - 3x - 4y - 4 = 0$ which passes through the origin such that the origin divides it in the ratio $4 : 1$.

(A) $3\mathrm{y} + 24\mathrm{x} = 0$

(B) $y = 0$

(C) $y = 5$

(D) $7y + 24x = 0$

Q.11

The range of values of 'a' such that angle θ between the pair of tangent drawn from $(a, 0)$ to the circle $x^2 + y^2 = 1$ satisfies $\frac{\pi}{2} < \theta < \pi$, lies in -

(A) $(1, 2)$

(B) $(1, \sqrt{2})$

(C) $(-\sqrt{2}, -1)$

(D) $(-2, -1)$

Q.12

Passage:
Consider a family of circles passing through the points $(3, 7)$ and $(6, 5)$.
Question:
Number of circles which belong to the family and also touching the X-axis are $-$

(A) 0

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(B) 1

(C) 2

(D) Infinite

Q.13

Passage:
Consider a family of circles passing through the points $(3, 7)$ and $(6, 5)$.
Question: If each circle in the family cuts the circle $x^2 + y^2 - 4x - 6y - 3 = 0$, then all the common chords pass through the fixed point which is -

(A) $(1, 23)$

(B) $(2, 23/2)$

(C) $(-3, 3/2)$

(D) None of these

Q.14

Passage:

Let S be the circle represented by the equation, $x^2 + y^2 - 8x - 10y + 37 = 0$
Question:

The least distance between the two points P and Q on the circles S and $S_1 \equiv x^2 + y^2 + 16x + 55 = 0$ is -

(A) 5 units

(B) 13 units

(C) 8 units

(D) None of these

Q.15

Paragraph
Consider the family of circles $x^2 + y^2 - 2x - 2\lambda y - 8 = 0$ passing through two fixed points A and B. Also $S = 0$ is a circle of this family, the tangent to which at A and B intersect on the line $x + 2y + 5 = 0$.**Question**
The distance between the points A and B, is

(A) 4

(B) $4\sqrt{2}$

(C) 6

(D) 8

Q.16

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Let AB be the chord of contact of the point $(5, -5)$ w.r.t the circle $x^2 + y^2 = 5$, then the locus of the orthocentre of ΔPAB where P is any point moving on the circle is

(A) $(x-1)^2 + (y+1)^2 = 5$

(B) $(x-1)^2 + (y+1)^2 = \frac{5}{2}$

(C) $(x+1)^2 + (y-1)^2 = 5$

(D) $(x+1)^2 + (y-1)^2 = \frac{5}{2}$

Q.17

Passage:

Let S be the circle represented by the equation, $x^2 + y^2 - 8x - 10y + 37 = 0$

Question:

If O is the origin and OP, OQ are the tangents from the origin to the circle S, the circumcentre of the triangle OPQ is -

(A) $\left(-2, \frac{5}{2}\right)$

(B) $\left(2, \frac{5}{2}\right)$

(C) $\left(-2, \frac{-5}{2}\right)$

(D) $\left(2, \frac{-5}{2}\right)$

Q.18

Match the column -

Column I	**Column II**
(A) If two circles $x^2 + y^2 + 2a_1x + b = 0$ and $x^2 + y^2 + 2a_2x + b$ touch each other then triplet (a_1, a_2, b) can be	(p) (2, 2, 2)
(B) If two circles $x^2 + y^2 + 2a_1x + b = 0$ and $x^2 + y^2 + 2a_2y + b$ touch each other then triplet (a_1, a_2, b) can be	(q) (1, 1, 1/2)
(C) If the straight line $a_1x - by + b^2 = 0$ touches the circle $x^2 + y^2 = a_2x + by$, then triplet (a_1, a_2, b) can be	(r) (2, 1, 0)
(D) If the line $3x + 4y - 4 = 0$ touches the circle $(x - a_1)^2 + (y - a_2)^2 = b^2$, then triplet (a_1, a_2, b) can be	(s) (1, 1, 3/5)

(A) (A) - q, (B) - p,q, (C) - r,s, (D) - p,q

(B) (A) - p, (B) - r,s, (C) - q,p, (D) - r,s

(C) (A) - p, (B) - q,s, (C) - s,q, (D) - p,r

(D) (A) - r, (B) - p,q, (C) - q,r, (D) - p,s

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Q.19

From a point P outside a circle with centre at C, tangents PA and PB are drawn such that $\frac{1}{(CA)^2}+\frac{1}{(PA)^2}=\frac{1}{16}$, then find the length of chord AB.

Q.20

Circle C_1 and C_2 touches externally and circles C_1 and C_2 touches internally to the circle C_3. The radii of C_1 and C_2 are 4 and 10 respectively and the centres of the three circles are collinear. A chord of C_3 is also a common transverse tangent of C_1 and C_2. Given that the length of the chord is $\frac{m\sqrt{n}}{p}$, where m, n and p are positive integers, m and p are relatively prime and n is not divisible by the square of any prime, then find the value of $\frac{(m+n+p)}{19}$.

Answers & Solutions

Q.1 Answer:

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Solution:

If $x^2+y^2+2gx+2fy+c=0$ is the circle which is orthogonal to the given three circles, the condition for orthogonal intersection gives the equations $-2g+3f=c-7, 5g-5f=c+9$, $7\,g-9f=c+29$,
which given the values $g=-8, f=-9$ and $c=-4$
The equation of the circle is $x^2+y^2-16x-18y-4=0$
Substituting $y=0, x^2-16x-4=0$
Length of the chord cut along the x-axis $=|$ difference between the roots of $x^2-16x-4=0\,|$ $=4\sqrt{17}$.

Q.2 Answer:

$(8,3);(14/13,-21/13)$

Solution:

The first circle, S_1, is $(x-3)^2+(y-3)^2=25$; its centre C_1 is $(3,3)$ and its radius is 5.
The second circle, S_2, is $(x-5)^2+y^2=18$; its centre C_2, is $(5,0)$ and its radius is $3\sqrt{2}$.
Subtracting the equations of the circles we obtain, after division by 2, $2x-3y-7=0$ ...(i)
Then p_1, the perpendicular distance from $C_1(3,3)$ to the line (i), is $10/\sqrt{13}$ which is less than $r_1(\equiv 5)$.
Similarly p_2, the perpendicular distance from $C_2(5,0)$ to (i), is $3/\sqrt{13}$ which is less than $r_2(\equiv 3\sqrt{2})$.
Accordingly, the circles intersect and (i) is the equation of the common chord AB.
The coordinates of A and B are obtained by solving (i) and one of the equations of the circles in this case the second equation is the simpler for this purpose.
On eliminating y between the two equations concerned we obtain $9x^2+(2x-7)^2-90x+63=0$ or $13x^2-118x+112=0$ or $(x-8)(13x-14)=0$.
Thus the abscissa of A and B are 8 and $14/13$; from (i) the corresponding ordinates are 3 and $-21/13$; A and B are the points $(8,3)$ and $(14/13,-21/13)$.

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Q.3 Answer:

$(4\sqrt{2}-\sqrt{14}, 5\sqrt{2})$

Solution:

The given point is an interior point

$\Rightarrow \left(-5+\frac{r}{\sqrt{2}}\right)^2+\left(-3+\frac{r}{\sqrt{2}}\right)^2-16<0 \Rightarrow r^2-8\sqrt{2}r+18<0$

$\Rightarrow 4\sqrt{2}-\sqrt{14}<r<4\sqrt{2}+\sqrt{14}.....(1)$

$\Rightarrow$ The point is on the major segment

$\Rightarrow$ The centre and the point are on the same side of the line $x+y=2$.

$\Rightarrow \quad -5+\frac{r}{\sqrt{2}}-3+\frac{r}{\sqrt{2}}-2<0$

$\Rightarrow \quad r<5\sqrt{2}....(2)$

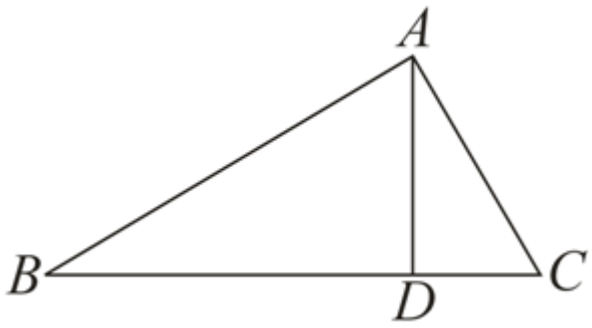
From eq. (1) and eq. (2), $4\sqrt{2}-\sqrt{14}<\mathrm{r}<5\sqrt{2}$

Q.4 Answer:

orthocenter

Solution:

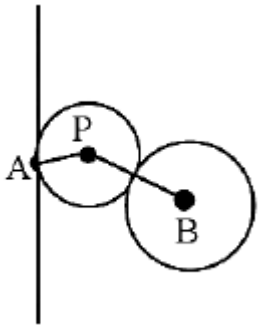
Since, $\angle ADB=\angle ADC=90^{\circ}$**, circle on** AB **and** AC **as diameters pass through** D **and therefore the altitude** AD **is the common chord. Similarly, the other two common chords are the other two altitudes and hence they concur at the orthocenter.**



Q.5 Answer:

$y^2-10x-6y+14=0$

Solution:



P **be** (h, k) **then** $\mathrm{PB}=2+\mathrm{h}$

$\Rightarrow (h-3)^2+(k-3)^2=h^2+4+4h$

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$\Rightarrow k^2 - 6k + 14 = 4h + 6h$

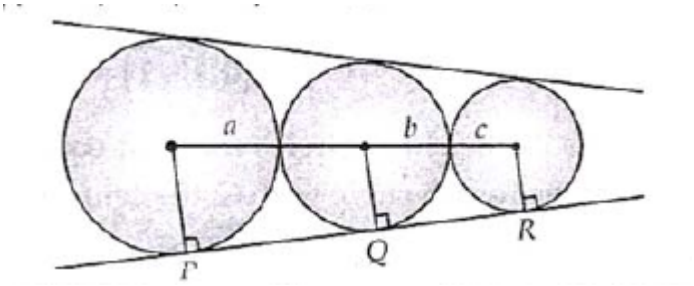
$\therefore$ **Locus of** P **is** $y^2 - 6y - 10x + 14 = 0$

Q.6 Answer:

$\sqrt{ac}$

Solution:

$PQ = \sqrt{(a+b)^2 - (a-b)^2} = 2\sqrt{ab}$



$QR = \sqrt{(b+c)^2 - (b-c)^2} = 2\sqrt{bc}, PR = \sqrt{(a+2b+c)^2 - (a-c)^2}$

$PQ + QR = PR \quad \therefore \quad \sqrt{ab} + \sqrt{bc} = \sqrt{b^2 + ac + b(a+c)}$

Squaring, $ab + bc + 2b\sqrt{ac} = b^2 + ac + b(a+c)$

$b^2 - 2b\sqrt{ac} + ac = 0 \Rightarrow (b - \sqrt{ac})^2 = 0$

$\therefore \quad b = \sqrt{ac}.$

Q.7 Answer:

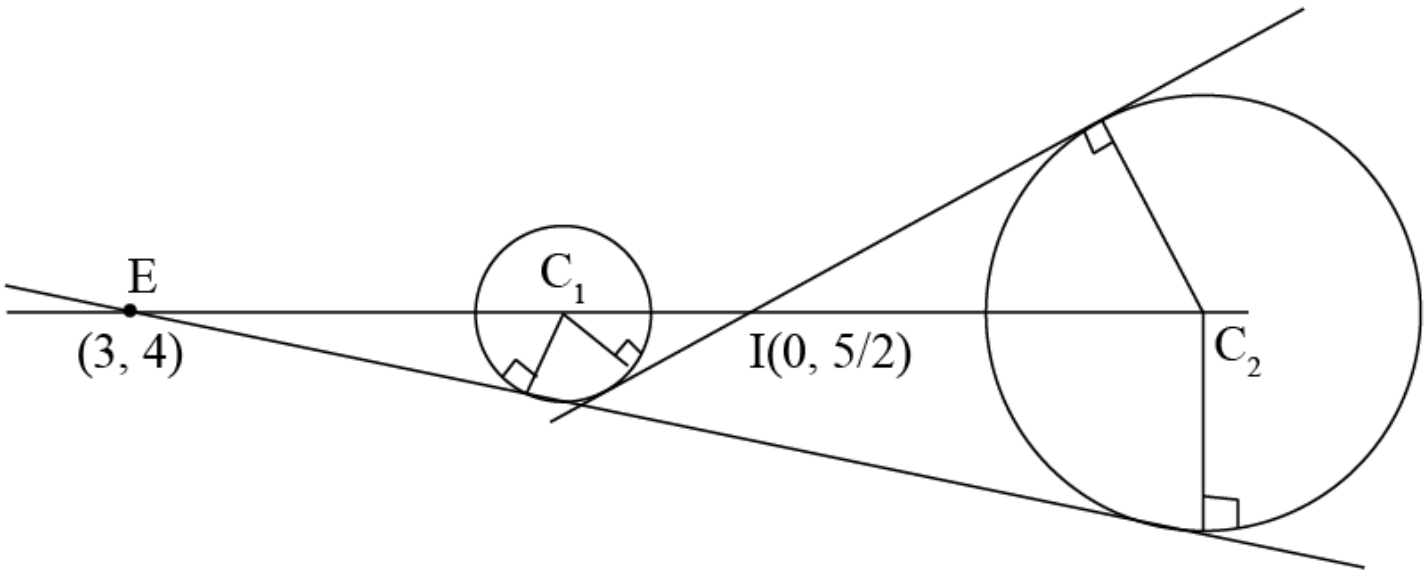
$\mathrm{y} = 4; 4\mathrm{x} - 3\mathrm{y} = 0$

Solution:

The centres are $C_1(1,3)$ **and** $C_2(-3,1)$ **respectively and the radii are** $r_1 = \sqrt{1+9-9} = 1$ **and** $r_2 = \sqrt{9+1-1} = 3$

point I dividing $\overline{C_1C_2}$ **internally in the ratio** $1:3$ **is** $\left(\frac{-3+3}{4} \cdot \frac{10}{4}\right)$ **or** $\left(0, \frac{5}{2}\right)$.

Similarly point E **dividing** $\overline{\mathrm{C_1C_2}}$ **externally in the ratio** $1:3$ **is** $(3,4)$.



The equation of a transverse common tangent is of the form $y - 5/2 = m(x-0)$ **or** $2mx - 2y + 5 = 0$

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Since this is tangent to the first circle, its distance from the centre $(1, 3)$ is same as its radius 1

So, $\frac{2\,m(1)-2(3)+5}{\sqrt{4\,m^2+4}} = 1$ **or,** $(2\,m-1)^2 = 4\,m^2+4$ **or,** $m = -\frac{3}{4}$

Since the two circles does not touch each other externally, $\therefore$ **second value of** $m =$ **infinity, so the equations of transverse common tangent are** $x = 0$ **and** $3x + 4y - 10 = 0$.

The equation of a direct common tangent is of form

$y - 4 = m(x - 3)$ or $mx - y + 4 - 3m = 0$

Since this line is tangent to the circle, we get

$\frac{m(1)-3+4-3\,m}{\sqrt{m^2+1}} = 1$ **or** $(1-2\,m)^2 = m^2+1$ **or** $(3\,m-4)m = 0$ **or** $m = 0, 4/3$

So the common tangents are $y - 4 = 0(x-3)$ **or** $y = 4$ **and** $y - 4 = \frac{4}{3}(x-3)$ **or** $4x - 3y = 0$

Q.8 Answer:

1

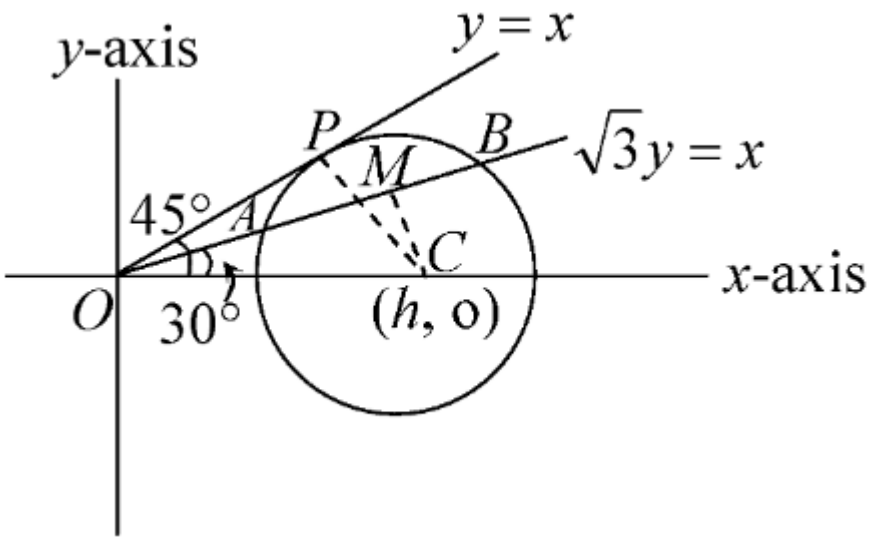
Solution:

$OP = CP = \frac{h}{\sqrt{2}} = AC$

$CM = h\sin 30^\circ = \frac{h}{2}$

From ΔACM

$AC^2 = AM^2 + CM^2$



$$\Rightarrow \quad \frac{h^2}{2} = 1 + \frac{h^2}{4} \Rightarrow \quad \frac{h^2}{4} = 1 \quad \Rightarrow \quad h = 2$$

$\therefore$ Radius $CP = \sqrt{2}$

$(x-2)^2 + y^2 = 2 \Rightarrow x^2 + y^2 - 4x + 2 = 0$

Q.9 Answer:

$\mathrm{OA}^2 + \mathrm{OB}^2 + \mathrm{BC}^2 \in [7, 11]$, where O is the origin

Solution:

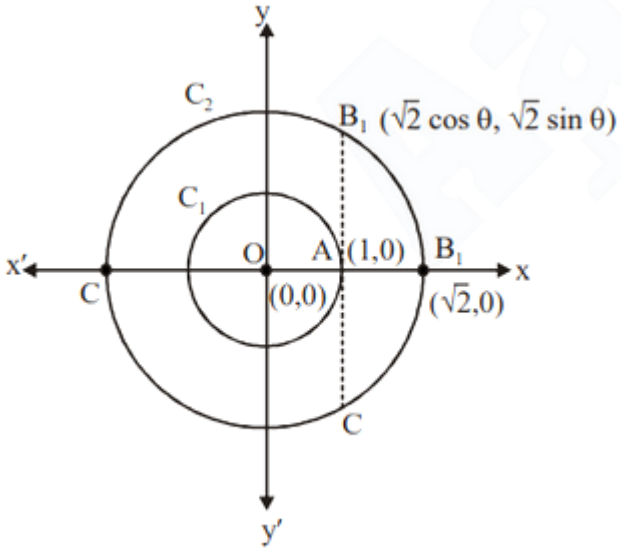
We have maximum $\mathrm{BC} = 2\sqrt{2}$ **and minimum** $\mathrm{BC} = 2$

$\therefore \mathrm{OA}^2 + \mathrm{OB}^2 + \mathrm{BC}^2 \in [7, 11]$

Let M **be the midpoint of** AB.

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$\Rightarrow M \equiv \left(\frac{1+\sqrt{2}\cos\theta}{2}, \frac{\sqrt{2}\sin\theta}{2}\right) = (h, k)$



$\therefore \quad \sin\theta = \frac{2k}{\sqrt{2}}, \cos\theta = \frac{2h-1}{\sqrt{2}}$

$\therefore$ **Now on squaring & adding, we get**

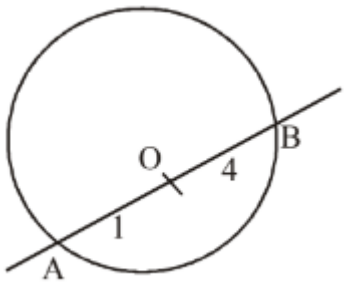
$\Rightarrow 4k^2 + (2h-1)^2 = 2$

$\therefore$ **Locus of** $M(h,k)$ **is** $\left(x-\frac{1}{2}\right)^2 + y^2 = \left(\frac{1}{\sqrt{2}}\right)^2$

Q.10 Answer:

$y = 0$

Solution:



$y = mx$ **be chord passing through** $A(x_1, y_1) \Rightarrow y_1 = mx_1$

$\Rightarrow x_1^2 + m^2x_1^2 - 3x_1 - 4mx_1 - 4 = 0$

$\Rightarrow (1+m^2)x_1^2 - (3+4m)x_1 - 4 = 0 \ldots.(1)$

$B(-4x, -4mx_1)$

$16x_1^2 + 16m^2x_1^2 + 12x_1 + 16mx_1 - 4 = 0$

$4(1+m^2)x_1^2 + (3+4m)x_1 - 1 = 0 \ldots.(2)$

Eq. (1) and (2) has common root

$x_1^2 = \frac{1}{1+m^2}$ **and** $5(3+4m)x_1 + 15 = 0 \quad ; x_1 = \frac{-3}{3+4m} \Rightarrow \frac{9}{(3+4m)^2} = \frac{1}{1+m^2}$

$9 + 9m^2 = 9 + 16m^2 + 24m$

$m = 0$ **or** $m = -24/7$

$y = 0$ **or** $7y + 24x = 0$

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Q.11 Answer:

$(1, \sqrt{2})$

Solution:

Equation of pair of tangents by $\mathrm{SS'} = \mathrm{T}^2$ **is** $(\mathrm{ax} + 0 - 1)^2 = \left(\mathrm{x}^2 + \mathrm{y}^2 - 1\right)\left(\mathrm{a}^2 + 0 - 1\right)$

or $\left(a^2 - 1\right) y^2 - x^2 + 2ax - a^2 = 0$

If θ **be the angle between the tangents, then**

$\tan\theta = \frac{2\sqrt{\mathrm{H}^2 - \mathrm{AB}}}{\mathrm{A}+\mathrm{B}} = \frac{2\sqrt{-(\mathrm{a}^2-1)(-1)}}{\mathrm{a}^2-2} = \frac{2\sqrt{\mathrm{a}^2-1}}{\mathrm{a}^2-2}$

If θ **lies in II quadrant, then** $\tan\theta < 0$

$\therefore \quad \frac{2\sqrt{a^2-1}}{a^2-2} < 0 \Rightarrow a^2 - 1 > 0$ **and** $a^2 - 2 < 0$

$\Rightarrow \quad |\mathrm{a}| > 1$ **and** $|\mathrm{a}| < 2$

$\Rightarrow \quad a \in (-\sqrt{2}, -1) \cup (1, \sqrt{2})$

Q.12 Answer:

2

Solution:

Equation of line passing through the points $A(3, 7)$ **and** $B(6, 5)$ **is** $y - 7 = -\frac{2}{3}(x - 3)$ **or** $2x + 3y - 27 = 0$

Also equation of circle with A and B as diameter end points is $(x - 3)(x - 6) + (y - 7)(y - 5) = 0$

Now family of circle through A and B is

$(x - 3)(x - 6) + (y - 7)(y - 5) + \lambda(2x + 3y - 27) = 0$ **....(1)**

If circle belonging to this family touches the x**-axis, then equation**

$(x - 3)(x - 6) + (0 - 7)(0 - 5) + \lambda(2x + 3(0) - 27) = 0$

has two equal roots, for which Discriminant $D = 0$**, which gives two values of** λ**.**

Equation of common chord of (1) **and** $x^2 + y^2 - 4x - 6y - 3 = 0$ **is radical axis, which is** $[(x - 3)(x - 6) + (y - 7)(y - 5) + \lambda(2x + 3y - 27)] - \left[x^2 + y^2 - 4x - 6y - 3\right] = 0$

or $\quad (2\lambda - 5)x + (3\lambda - 6)y + (-27\lambda + 56) = 0$

or $(-5x - 6y + 56) + \lambda(2x + 3y - 27) = 0.$

This is family of lines which passes through the point of intersection of $(-5x - 6y + 56 = 0$ **and** $2x + 3y - 27 = 0$**, which is** $(2, 23/3)$

If circle (1) cuts $x^2 + y^2 = 29$ **orthogonally, then**

$0 + 0 = -29 + 56 - 27\lambda = 0$

$\Rightarrow \lambda = 1$

$\Rightarrow$ **Required circle is** $x^2 + y^2 - 7x - 9y + 26 = 0$**, centre is** $(7/2, 9/2)$**.**

Q.13 Answer:

None of these

Solution:

Equation of line passing through the points $A(3, 7)$ **and** $B(6, 5)$ **is** $y - 7 = -\frac{2}{3}(x - 3)$ **or** $2x + 3y - 27 = 0$

Also equation of circle with A and B as diameter end points is $(x - 3)(x - 6) + (y - 7)(y - 5) = 0$

Now family of circle through A and B is

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$(x-3)(x-6)+(y-7)(y-5)+\lambda(2x+3y-27)=0$ ....(1)

If circle belonging to this family touches the x-axis, then equation

$(x-3)(x-6)+(0-7)(0-5)+\lambda(2x+3(0)-27)=0$

has two equal roots, for which Discriminant $D=0$, which gives two values of λ.

Equation of common chord of (1) and $x^2+y^2-4x-6y-3=0$ is radical axis, which is $[(x-3)(x-6)+(y-7)(y-5)+\lambda(2x+3y-27)]-\left[x^2+y^2-4x-6y-3\right]=0$

or $(2\lambda-5)x+(3\lambda-6)y+(-27\lambda+56)=0$

or $(-5x-6y+56)+\lambda(2x+3y-27)=0.$

This is family of lines which passes through the point of intersection of $(-5x-6y+56=0$ and $2x+3y-27=0$, which is $(2,23/3)$

If circle (1) cuts $x^2+y^2=29$ orthogonally, then

$0+0=-29+56-27\lambda=0$

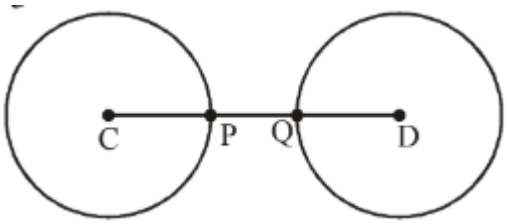
$\Rightarrow \lambda=1$

$\Rightarrow$ **Required circle is $x^2+y^2-7x-9y+26=0$, centre is $(7/2,9/2)$.**

Q.14 Answer:

8 units

Solution:



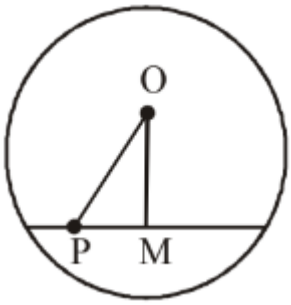
Centre of S **is** $\mathrm{C}(4,5)$ **and** S_1 **is** $\mathrm{D}(-8,0)$**. Their radius** $\mathrm{r}_1=2$ **and** $\mathrm{r}_2=3$**.** P **and** Q **lie on the circles. Hence,** PQ **will be least when** $\mathrm{C},\mathrm{P},\mathrm{Q},\mathrm{D}$ **are collinear.**

$\therefore$ **Least value of** $PQ=CD-(r_1+r_2)=\sqrt{12^2+5^2}-(2+3)=8$ **units**

(ii) Let $\mathrm{P}(2,3)$ **be given point,** M **be the middle point of a chord of the circle** $x^2+y^2-8x-10y+37=0$ **through** P**.**

$(\mathrm{OM})^2=\mathrm{OP}^2-\mathrm{PM}^2$ **which is maximum when** PM **is minimum.**

i.e., P **coincides with** M**.**



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Equation of chord is $\mathrm{T} = \mathrm{S}'$.

i.e., $2x + 3y - 4(x + 2) - 5(y + 3) + 37 = 0$

$\Rightarrow 2x + 2y - 14 = 0$
$\Rightarrow x + y - 7 = 0$

(iii) We note that PQ is the chord of contact of the tangents from the origin to the circle

$x^2 + y^2 - 8x - 10y + 37 = 0$ **.....(1)**

Equation of PQ is

$4x + 5y - 37 = 0$ **.....(2)**

Equation of a circle passing through the intersection of (1) and (2) is

$$x^2 + y^2 - 8x - 10y + 37 + \lambda(4x + 5y - 37) = 0 \quad \ldots\ldots\ldots\ldots\ldots (3)$$

If this represents the circumcircle of the triangle OPQ**, it passes through** $\mathrm{O}(0, 0)$**, so from (3),** $\lambda = 1$

$\therefore$ **equation (3) becomes** $x^2 + y^2 - 4x - 5y = 0$

So, the required coordinates of the centre are $(2, 5/2)$**.**

Q.15 Answer:

6

Solution:

We have $\left(x^2 + y^2 - 2x - 8\right) - 2\lambda y = 0$**, which is the form of** $S + \lambda L = 0$
On solving $S = 0$ **and** $L = 0$ **where** $S \equiv x^2 + y^2 - 2x - 8 = 0$ **and** $L \equiv y = 0$
Put $y = 0 \Rightarrow x^2 - 2x - 8 = 0 (x - 4)(x + 2) = 0$
$x = -2$ **or 4**
$A \equiv (4, 0); B \equiv (-2, 0)$
Distance between A **and** $B = 6$

Q.16 Answer:

$(x - 1)^2 + (y + 1)^2 = 5$

Solution:

Equation of chord of contact AB **is**

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$5x - 5y = 1$

$\Rightarrow x - y = 1$

Therefore,

$x^2 + (x-1)^2 = 5$

$\Rightarrow 2x^2 - 2x - 4 = 0$

$\Rightarrow x = 2, \ -1$

$\Rightarrow y = 1, \ -2$

Hence, $A(2,1)$ **and** $B(-1,-2)$**.**

And

$P \equiv \left(\sqrt{5}\cos\theta, \sqrt{5}\sin\theta\right)$

Let the locus of orthocenter be (h,k)**. Now, we know that the centroid divides the line joining orthocenter and circumcenter in** $2:1$**. Therefore,**

$h = 2 - 1 + \sqrt{5}\cos\theta = 1 + \sqrt{5}\cos\theta$

$k = 1 + (-2) + \sqrt{5}\sin\theta = -1 + \sqrt{5}\sin\theta$

$\Rightarrow (h-1)^2 + (k+1)^2 = 5$

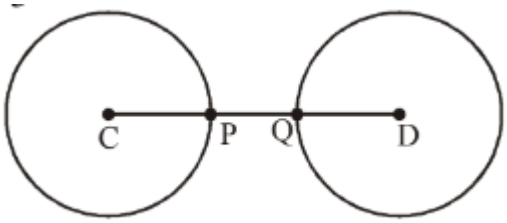
Hence, the required locus is

$(x-1)^2 + (y+1)^2 = 5$

Q.17 Answer:

$\left(2, \frac{5}{2}\right)$

Solution:



Centre of S **is** $\mathrm{C}(4,5)$ **and** S_1 **is** $\mathrm{D}(-8,0)$**.**

Their radius $\mathrm{r}_1 = 2$ **and** $\mathrm{r}_2 = 3$**.** P **and** Q **lie on the circles.**

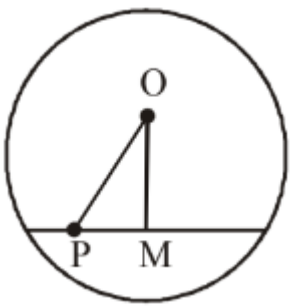
Hence, PQ **will be least when** C, P, Q, D **are collinear.**

$\therefore$ **Least value of** $PQ = CD - (r_1 + r_2) = \sqrt{12^2 + 5^2} - (2+3) = 8$ **units**

(ii) Let $\mathrm{P}(2,3)$ **be given point,** M **be the middle point of a chord of the circle** $x^2 + y^2 - 8x - 10y + 37 = 0$ **through** P**.**

$(\mathrm{OM})^2 = \mathrm{OP}^2 - \mathrm{PM}^2$ **which is maximum when** PM **is minimum.**

i.e., P **coincides with** M**.**



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Equation of chord is $T = S'$.

i.e., $2x + 3y - 4(x + 2) - 5(y + 3) + 37 = 0$

$\Rightarrow 2x + 2y - 14 = 0$

$\Rightarrow x + y - 7 = 0$

(iii) We note that PQ is the chord of contact of the tangents from the origin to the circle

$x^2 + y^2 - 8x - 10y + 37 = 0$ **.....(1)**

Equation of PQ is

$4x + 5y - 37 = 0$ **.....(2)**

Equation of a circle passing through the intersection of (1) and (2) is

$x^2 + y^2 - 8x - 10y + 37 + \lambda(4x + 5y - 37) = 0 \quad \ldots\ldots\ldots\ldots\ldots (3)$

If this represents the circumcircle of the triangle OPQ**, it passes through** $O(0, 0)$**, so from (3),** $\lambda = 1$

$\therefore$ **equation (3) becomes** $x^2 + y^2 - 4x - 5y = 0$

So, the required coordinates of the centre are $(2, 5/2)$.

Q.18 Answer:

(A) - r, (B) - p,q, (C) - q,r, (D) - p,s

Solution:

$(A) - r, (B) - pq, (C) - qr, (D) - ps$

(A) Radical axis of $x^2 + y^2 + 2a_1x + b = 0$ **and** $x^2 + y^2 + 2a_2x + b$ **is** $(a_1 - a_2)\,x = 0$ **or** $x = 0$**, it must touch both of the circle.**

Solving it with one of the circles we get $y^2 + b = 0 \Rightarrow b \leq 0$

(B) Radical axis of $x^2 + y^2 + 2a_1x + b = 0$ **and** $x^2 + y^2 + 2a_2y + b = 0$ **is** $a_1x - a_2y = 0$

Solving it with one of the circles we have

$x^2 + (a_1/a_2)^2 x^2 + 2a_1x + b = 0$

This equation must have equal roots.

Hence, $4a_1^2 - 4\,b\left[1 + \left(a_1^2/a_2^2\right) = 0\right.$

$\Rightarrow \quad a_1^2 - b\left[1 + \left(a_1^2/a_2^2\right) = 0\right.$

Options p and q satisfy this condition.

(C) If the straight line $a_1x - by + b^2 = 0$ **touches the circle** $x^2 + y^2 = a_2x + by$

$\Rightarrow \quad \frac{\left|a_1\frac{a_2}{2} - b\frac{b}{2} + b^2\right|}{\sqrt{a_1^2 + b^2}} = \sqrt{\frac{a_2^2}{4} + \frac{b^2}{4}} \Rightarrow \frac{\left|a_1a_2 + b^2\right|}{\sqrt{a_1^2 + b^2}} = \sqrt{a_2^2 + b^2}$

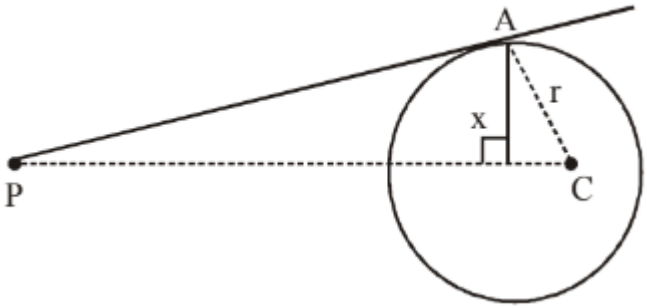
$\Rightarrow \quad a_1^2a_2^2 + 2\,b^2a_1a_2 + b^4 = a_1^2a_2^2 + a_1^2\,b^2 + a_2^2\,b^2 + b^4$

$\Rightarrow \quad b^2 = 0 \text{ or } 2a_1a_2 = a_1^2 + a_2^2$

$\Rightarrow$ Option q and r

(D) Distance $3x + 4y - 4 = 0$ **touches the circle** $(x - a_1)^2 + (y - a_2)^2 = b^2$

$\Rightarrow \quad \frac{|3a_1 + 4a_2 - 4|}{5} = b$



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Q.19 Answer:

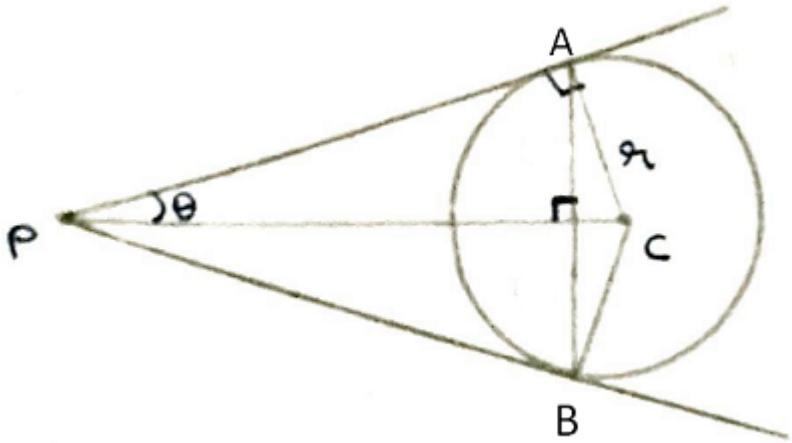
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Solution:

$\tan\theta = \frac{r}{PA}$

Given, $\frac{1}{r^2} + \frac{1}{PA^2} = \frac{1}{16} \Rightarrow \frac{\cot^2\theta+1}{(PA)^2} = \frac{1}{16}$

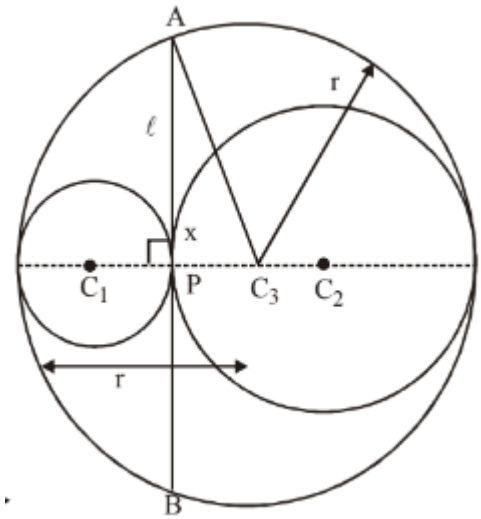
$\Rightarrow PA\sin\theta = 4 = x \Rightarrow 2x = 8$



Q.20 Answer:

—

Solution:



$r - 4 = C_1C_3$

$r - 10 = C_2C_3$

$2r - 14 = C_1C_2 = 10 + 4 = 14 \Rightarrow r = 14$

Now, $PC_3 = x = 14 - 8 = 6$

$\Rightarrow \quad \ell^2 = r^2 - x^2 = 14^2 - 6^2 = 160 \Rightarrow \ell = 4\sqrt{10}$

$2\ell = \frac{8\sqrt{10}}{1} \Rightarrow m = 8, n = 10, p = 1$

$\therefore \quad m + n + p = 19$

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